KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

EXPLORATORY DATA ANALYSIS AND DEEP LEARNING-BASED

Loan Approval Prediction

Assignment II

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Abstract

This project explores a Loan Approval Prediction dataset and develops a machine learning classification model to predict loan approval status. The analysis uses Python, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn. The dataset contains 58,645 records and 13 original columns related to applicant age, income, employment, home ownership, loan amount, interest rate, credit history, and loan status. Data preprocessing includes checking missing values, identifying duplicate records, encoding categorical variables, and preparing numerical features for model training. Descriptive visualizations examine loan amount distribution, applicant age, loan approval status, correlations, annual income versus loan amount, loan purpose, and home ownership. The classification model achieved 94.57% accuracy, 92.16% precision, 67.60% recall, an F1-score of 78.00%, and a ROC-AUC of 0.9290 on the recorded test evaluation. The results demonstrate the potential of machine learning to identify patterns associated with loan approval, although historical associations do not establish causation.

1. **1. Introduction and Objectives**

   Loan approval is an important process in financial services. Evaluating applicants involves considering factors such as income, employment history, credit history, loan amount, and the purpose of borrowing. Analyzing these factors can help identify patterns associated with loan approval outcomes. The objective of this project is to perform exploratory data analysis (EDA), visualize important patterns, preprocess the dataset, and develop a machine learning classification model to predict loan approval status.

   Objectives:

   - Inspect the dataset and summarize its important features.
   - Check missing values, duplicate records, and data types.
   - Analyze numerical and categorical variables.
   - Create visualizations and interpret the observed patterns.
   - Train and evaluate a machine learning classification model.
   - Assess model performance using accuracy, precision, recall, F1-score, and ROC-AUC.
   - Discuss the limitations of the model and possible future improvements.

1. **Dataset Description**

The dataset used in this project is a Loan Approval Prediction dataset obtained from Kaggle. It contains 58,645 records and 13 original columns. The dataset includes applicant age, annual income, home ownership, employment length, loan intent, loan grade, loan amount, interest rate, loan percentage of income, previous default information, credit history length, and loan status. The target variable is loan_status, where 0 represents a loan not approved and 1 represents a loan approved, according to the current project interpretation.

2. **3. EDA and Preprocessing**

The dataset was loaded using Pandas. The notebook was used to inspect column names, data types, descriptive statistics, missing values, and duplicate records. The initial inspection showed zero missing values and zero duplicate rows in the loaded dataset. Numerical and categorical features were examined to understand their distributions and identify patterns.

The target variable was analyzed to understand the distribution of loan outcomes. The dataset contains 50,295 records in class 0 and 8,350 records in class 1, indicating class imbalance. Categorical variables were encoded, and numerical features were prepared for machine learning. The data was divided into training and testing sets, and preprocessing was applied before model evaluation.

The notebook reports 14 input features after preprocessing, including the additional encoded columns. Data preparation and model evaluation were performed to support the prediction of loan approval status.

2. Seven Stages of Visualization

The visualization workflow in this project follows seven practical stages:

1. **Define the question:** Understand the factors associated with loan approval.
2. **Collect data:** Obtain the loan approval dataset from Kaggle.
3. **Prepare data:** Inspect data types, missing values, duplicates, and feature formats.
4. **Choose chart types:** Select histograms, count plots, scatter plots, grouped bar charts, and a correlation heatmap.
5. **Create charts:** Generate visualizations using Matplotlib and Seaborn.
6. **Interpret findings:** Examine distributions, relationships, and differences between loan status categories.
7. **Communicate results:** Present the charts and explain the findings in the project report.

3. Data Visualizations and Interpretations

5.1 Loan Approval Status Distribution

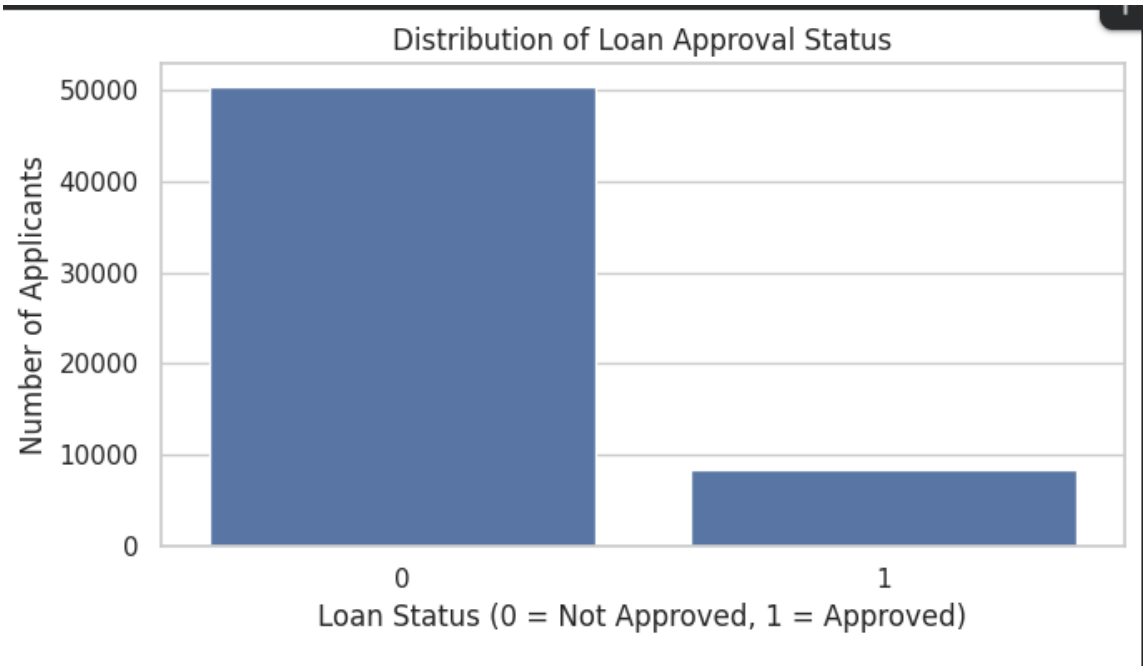


Figure 1. Distribution of loan approval status.

What it shows: The chart displays the number of applicants in each loan status category.

Why created: To understand the target distribution before building the machine learning model.

Insight: Class 0 contains 50,295 records, while class 1 contains 8,350 records. The classes are imbalanced, so accuracy should be considered alongside precision, recall, F1-score, and ROC-AUC.

5.2Booking Lead-Time Distribution

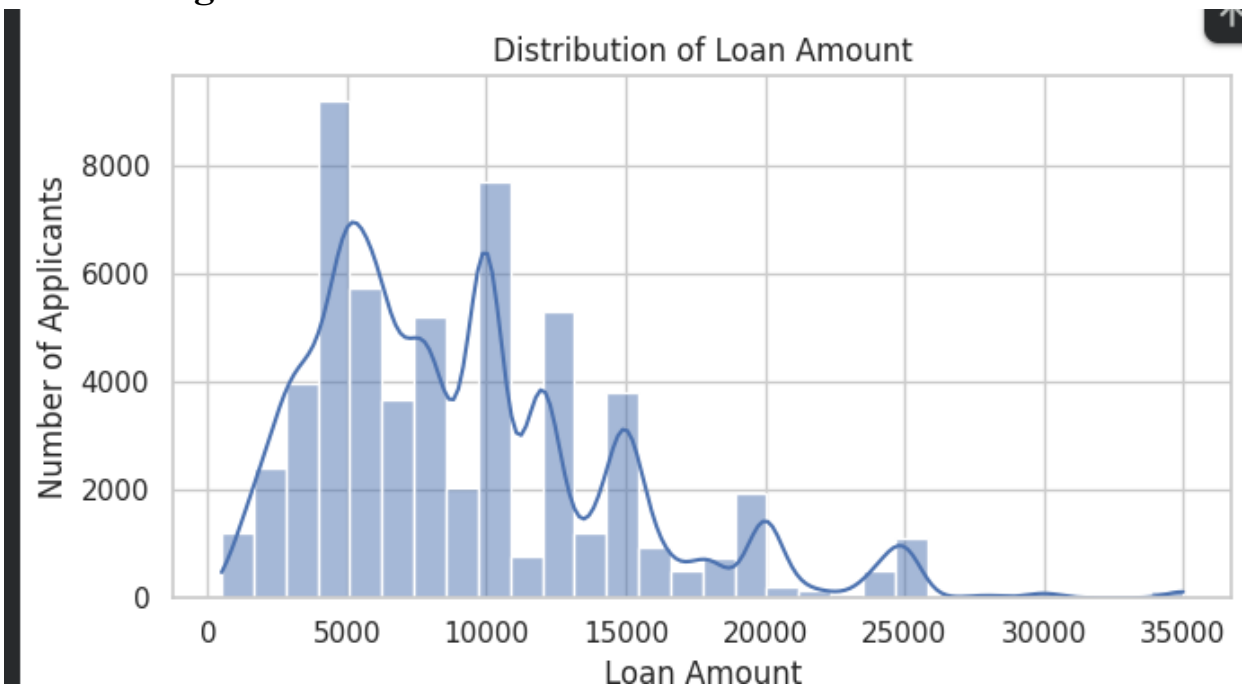


Figure 2. Distribution of loan amounts.

What it shows: The histogram illustrates how requested loan amounts are distributed among applicants.

Why created: To identify common loan amount ranges and observe the distribution of high-value requests.

Insight: Most observations occur within particular loan amount ranges, while fewer applicants request very large amounts.

5.3Distribution of Applicant Age

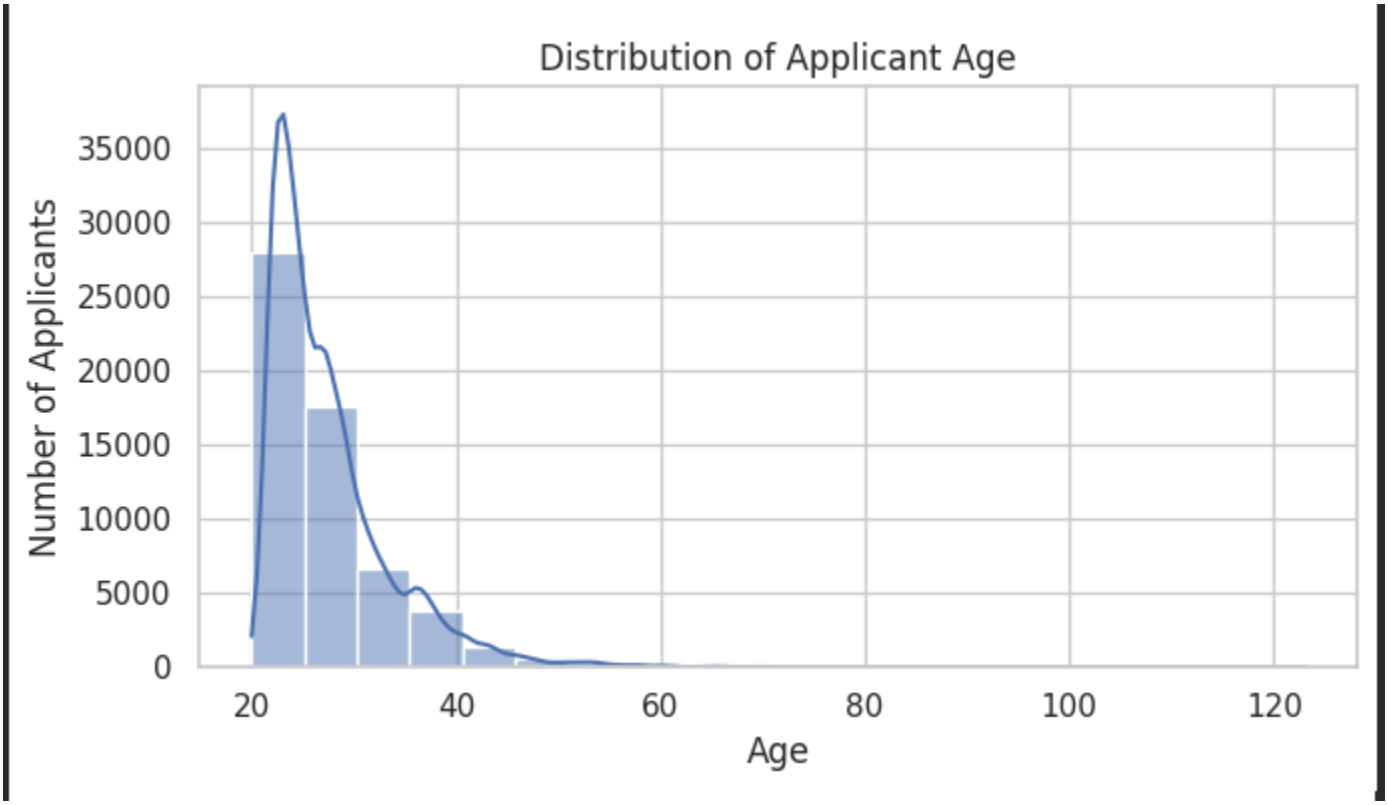


Figure 3. Distribution of applicant age.

What it shows: The histogram presents the age distribution of applicants.

Why created: To understand the demographic distribution and identify unusual age values that may require further investigation.

Insight: Applicant ages are concentrated in younger age groups, with fewer observations at higher ages. Extreme values should be checked before drawing conclusions.

. 5.4 Correlation Heatmap

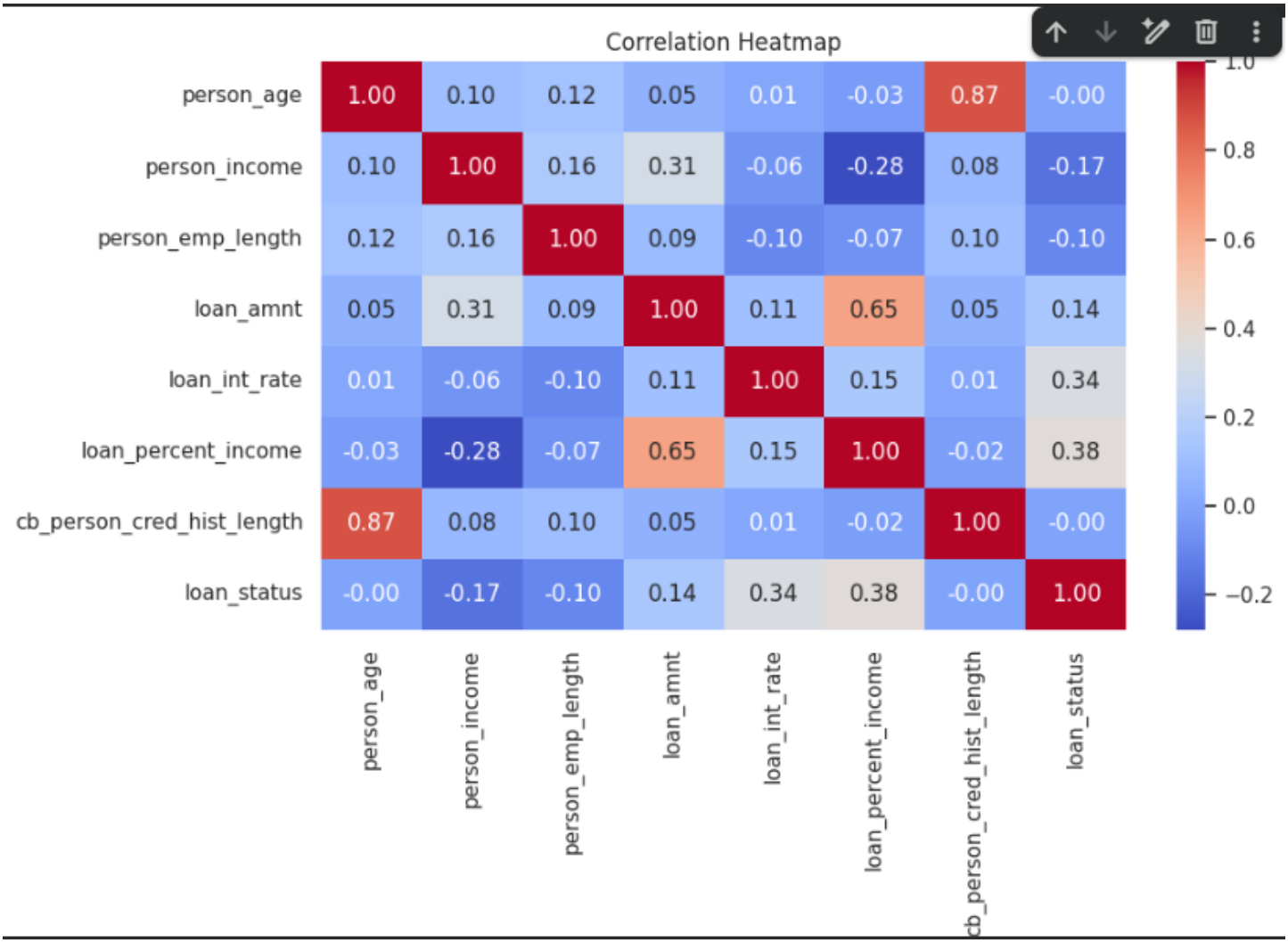


Figure 4. Correlation heatmap of selected numerical features.

What it shows: The heatmap illustrates correlations between numerical variables, including income, loan amount, interest rate, and loan status.

Why created: To explore relationships between numerical features and identify variables that may be useful for further analysis.

Insight: Some numerical variables show stronger associations than others. Correlation indicates a relationship between variables but does not establish causation.

5.5 Annual Income versus Loan Amount

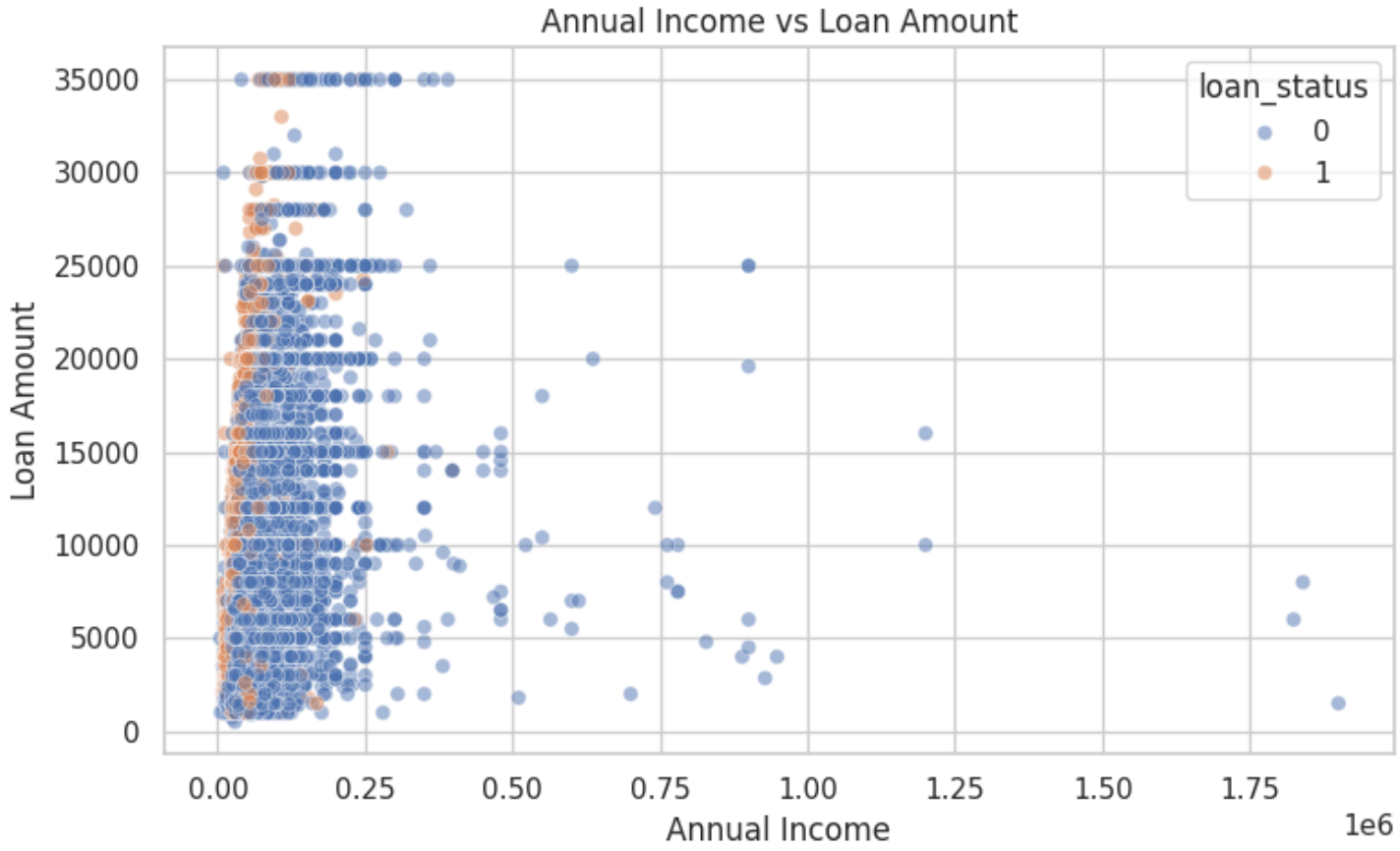


Figure 5. Annual income versus loan amount by loan status.

What it shows: The scatter plot compares annual income with requested loan amounts, distinguishing the two loan status classes.

Why created: To examine how requested amounts vary across income levels and whether the two classes show visible separation.

Insight: The observations overlap across income and loan amount ranges, suggesting that these two variables alone do not fully distinguish loan outcomes.

5.6 Loan Status by Loan Intent

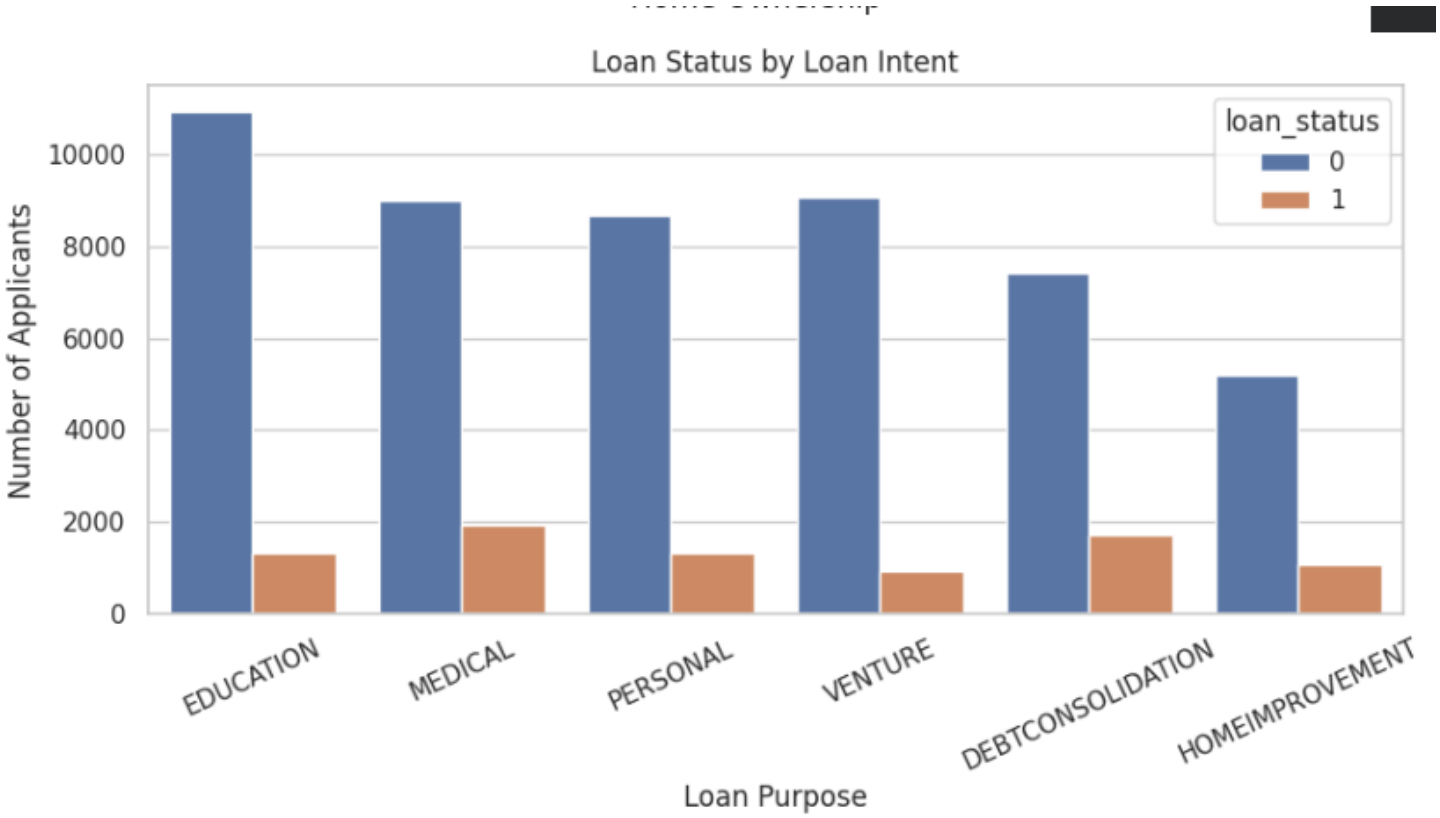


Figure 6. Loan status by loan intent.

What it shows: The grouped bar chart compares the number of applicants in each loan status category across different loan purposes.

Why created: To examine how loan outcomes vary across education, medical, personal, venture, debt consolidation, and home improvement loans.

Insight: The counts differ across loan purposes. Approval rates should be calculated within each category before comparing the likelihood of approval.

5.7 Loan Status by Home Ownership

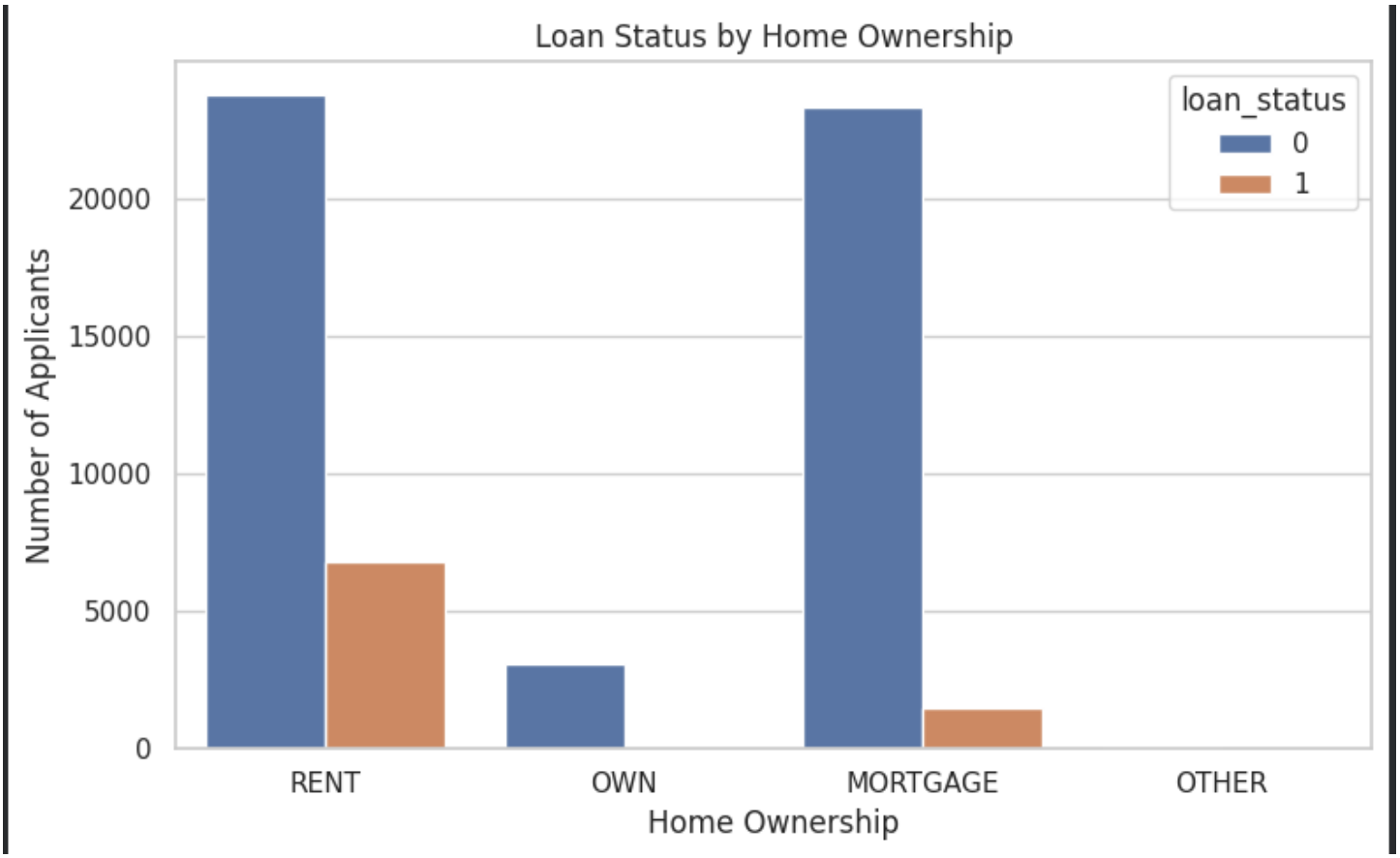


Figure 7. Loan status by home ownership.

What it shows: The chart compares loan status across home ownership categories.

Why created: To explore whether applicant home ownership categories are associated with different loan outcomes.

Insight: The number of observations differs across home ownership categories. These differences describe the dataset and do not prove that home ownership causes a particular loan decision.

4. Machine Learning Model

A machine learning classification model was developed to predict loan approval status using the Loan Approval Prediction dataset. The model uses applicant-related features such as age, annual income, employment length, home ownership, loan intent, loan amount, interest rate, and credit history.

The dataset was preprocessed before model training. Categorical variables were encoded, and numerical features were prepared for the classification process. The data was divided into training and testing sets to evaluate the model on unseen records.

The model's performance was evaluated using accuracy, precision, recall, F1-score, a confusion matrix, and the ROC curve. These metrics help assess the model's predictive performance and its ability to distinguish between the two loan status classes.

4. **5. Model Evaluation and Results**

The model was evaluated using the recorded test results.

Metric	Test Result
Accuracy	94.57%
Precision	92.16%
Recall	67.60%
F1-score	78.00%
ROC-AUC	0.9290

Accuracy of 94.57% means that approximately 94.57% of test predictions were correct. Precision of 92.16% indicates that most cases predicted as class 1 belonged to the actual positive class. Recall of 67.60% indicates that the model identified

approximately 67.60% of actual positive cases. The F1-score of 78.00% summarizes the balance between precision and recall. The ROC-AUC score of 0.9290 indicates good discrimination between the two classes.

5.1 Confusion Matrix

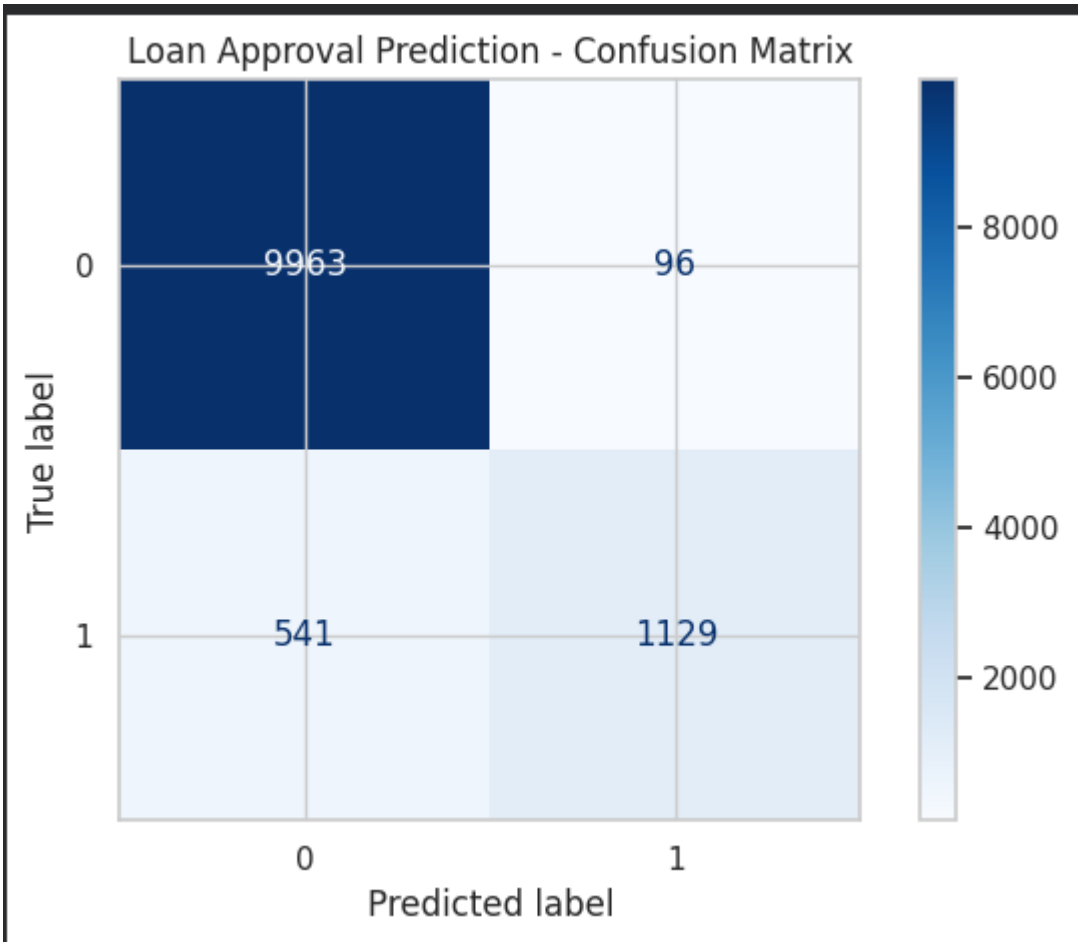


Figure 8. Confusion matrix for loan approval prediction.

The confusion matrix compares the actual loan status with the model's predicted status. It helps identify correct predictions and errors for both classes. In particular, false positives and false negatives should be examined to understand the risks associated with incorrect loan status predictions.

Note: Use the confusion matrix from the same test run as the reported evaluation metrics.

5.2 ROC Curve

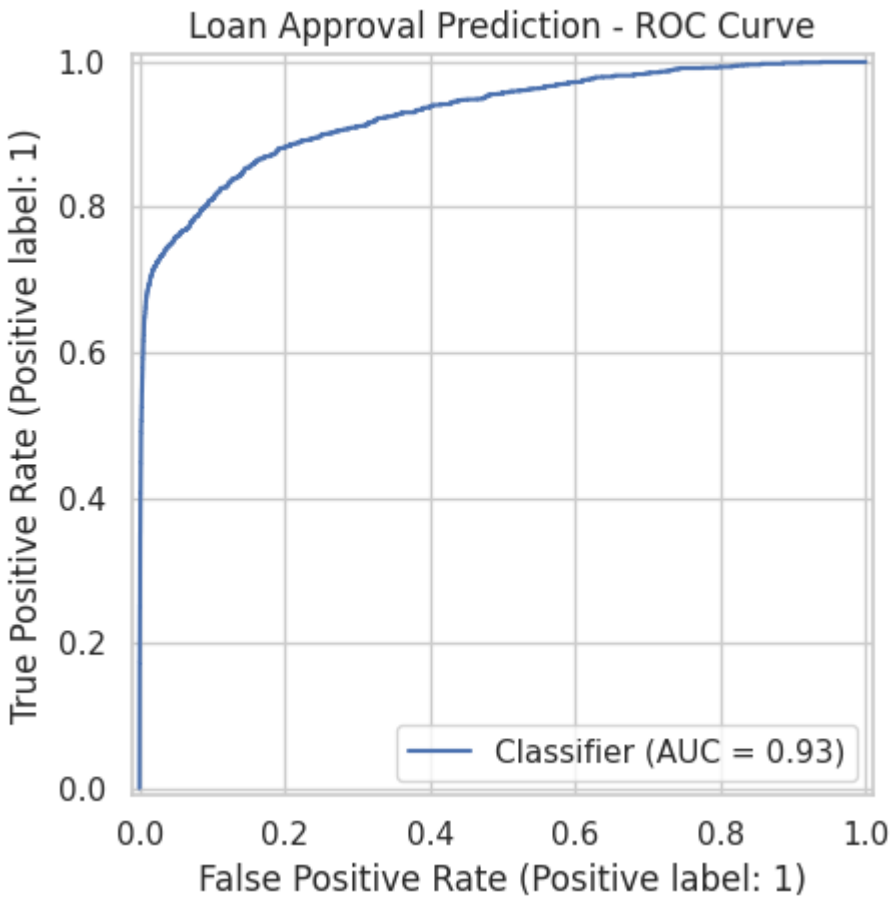


Figure 9. ROC curve for loan approval prediction.

The ROC curve illustrates the relationship between the true-positive rate and false-positive rate at different classification thresholds. The recorded ROC-AUC score of 0.9290 indicates that the model has good ability to distinguish between the two classes. However, this score does not mean that 92.90% of individual predictions are correct.

6. Conclusion and Future Scope

This project applied exploratory data analysis, statistical summaries, data visualization, and machine learning to a Loan Approval Prediction dataset containing 58,645 records and 13 original columns. The visualizations helped examine loan amount distribution, applicant age, loan status, numerical correlations, annual income versus loan amount, loan intent, and home ownership.

The recorded model evaluation achieved 94.57% accuracy, 92.16% precision, 67.60% recall, an F1-score of 78.00%, and a ROC-AUC of 0.9290. Although the model demonstrated good overall performance, its recall indicates that some actual positive cases were missed. The imbalance between the two target classes should also be considered when interpreting the results.

Future work could compare additional classification algorithms, use cross-validation, investigate class-imbalance handling techniques, tune the classification threshold, and evaluate feature importance. Since loan decisions affect individuals financially, the model should not be used as the sole basis for real lending decisions.

recall was 67.40%, meaning some actual cancellations were missed. Therefore, the model may help identify cancellation risk but should not be used as the only basis for hotel decisions.

Limitations include reliance on historical data, possible changes in customer behaviour and hotel policies, class imbalance, and the fact that correlations do not establish cause and effect. Future work could compare the MLP with tree-based models, tune the decision threshold according to business costs, test on newer data, and monitor model performance over time.

7. References

1. Kaggle. (n.d.). *Loan Approval Prediction dataset.* https://www.kaggle.com/
2. Pandas. (n.d.). *Pandas documentation.* https://pandas.pydata.org/docs/
3. Matplotlib. (n.d.). *Matplotlib documentation.* https://matplotlib.org/stable/
4. Seaborn. (n.d.). *Seaborn documentation.* https://seaborn.pydata.org/
5. Scikit-learn. (n.d.). *Scikit-learn documentation.* https://scikit-learn.org/stable/

8. Appendix — Source Code

The complete Python source code and generated outputs are maintained in the project's GitHub repository. The repository should include the Jupyter notebook, project report, README file, dataset or its source link, and generated output figures.

GitHub Repository URL:
https://github.com/bhavansri2006/Loan-Approval-Prediction